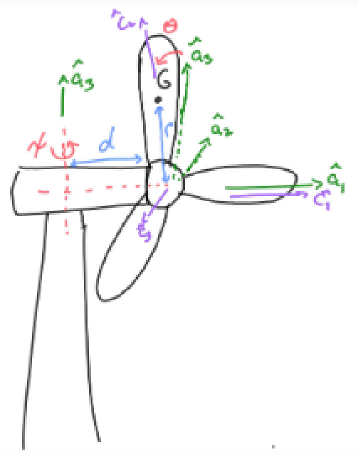
Problem 1

Problem 1

The rotor of a wind turbine is rotating by an angle θ about its center (or hub). The hub is offset from the centerline of the tower by a distance d along the "nacelle," which houses the gearing and electrical components of the wind turbine. The nacelle is allowed to yaw (rotate around the $\hat{a}_3$-direction) around the tower to align the wind turbine with the wind direction. This rotation is the angle ψ relative to some initial direction. If the rotors are spinning while the system is rotating in the yaw direction, what is the velocity and acceleration of the center of mass of one of the turbine blades, located at a distance r from the center of the rotor?



Reference Frames

$$\hat{a}_3 = \hat{n}_3, \qquad \vec{\omega}^{A/N} = \dot{\psi}\,\hat{a}_3, \qquad \hat{c}_1 = \hat{a}_1, \qquad \vec{\omega}^{C/A} = \dot{\theta}\,\hat{a}_1$$
$$\vec{\omega}^{C/N} = \vec{\omega}^{A/N} + \vec{\omega}^{C/A} = \dot{\theta}\,\hat{a}_1 + \dot{\psi}\,\hat{a}_3$$

Position of the Blade Center of Mass

$$\vec{r}_{G/O} = d\,\hat{a}_1 + r\,\hat{c}_2, \qquad \begin{Bmatrix}\hat{c}_1\\ \hat{c}_2\\ \hat{c}_3\end{Bmatrix} = \underbrace{\begin{bmatrix}1 & 0 & 0\\ 0 & \cos\theta & \sin\theta\\ 0 & -\sin\theta & \cos\theta\end{bmatrix}}_{[R]^{A/C}} \begin{Bmatrix}\hat{a}_1\\ \hat{a}_2\\ \hat{a}_3\end{Bmatrix}$$

$$\Rightarrow \qquad \vec{r}_{G/O} = d\,\hat{a}_1 + r\cos\theta\,\hat{a}_2 + r\sin\theta\,\hat{a}_3$$

Velocity of G

Applying the transport theorem between the C and N frames, with $\frac{^C d}{dt}\vec{r}_{G/O} = 0$:

$$^N\vec{v}_G = \frac{^C d}{dt}\vec{r}_{G/O} + \vec{\omega}^{C/N} \times \vec{r}_{G/O} = \vec{\omega}^{C/N} \times \vec{r}_{G/O}$$

$$^N\vec{v}_G = \begin{vmatrix}\hat{a}_1 & \hat{a}_2 & \hat{a}_3\\ \dot{\theta} & 0 & \dot{\psi}\\ d & r\cos\theta & r\sin\theta\end{vmatrix}$$

Expanding the determinant:

$$ {}^N\vec{v}_G = (0 \cdot r\sin\theta - \dot\psi\, r\cos\theta)\hat a_1 - (\dot\theta\, r\sin\theta - \dot\psi\, d)\hat a_2 + (\dot\theta\, r\cos\theta - 0\cdot d)\hat a_3 $$

$$ \boxed{{}^N\vec{v}_G = -r\dot\psi\cos\theta\;\hat a_1 + (\dot\psi\, d - r\dot\theta\sin\theta)\hat a_2 + r\dot\theta\cos\theta\;\hat a_3} $$

Acceleration of G

Applying the transport theorem between the A and N frames, with $\vec\omega^{A/N} = \dot\psi\,\hat a_3$:

$$ {}^N\vec{a}_G = \frac{{}^A d}{dt}\Big({}^N\vec{v}_G\Big) + \vec\omega^{A/N} \times {}^N\vec{v}_G $$

Term I: A-frame derivative of $\vec v_G$. Differentiating each scalar component (treating $\hat a_1, \hat a_2, \hat a_3$ as constant):

$$ \begin{aligned} \frac{d}{dt}\big(-r\dot\psi\cos\theta\big) &= -r\ddot\psi\cos\theta + r\dot\psi\dot\theta\sin\theta \\ \frac{d}{dt}\big(\dot\psi\, d - r\dot\theta\sin\theta\big) &= \ddot\psi\, d - r\ddot\theta\sin\theta - r\dot\theta^2\cos\theta \\ \frac{d}{dt}\big(r\dot\theta\cos\theta\big) &= r\ddot\theta\cos\theta - r\dot\theta^2\sin\theta \end{aligned} $$

$$ \frac{{}^A d}{dt}\big({}^N\vec{v}_G\big) = (r\dot\psi\dot\theta\sin\theta - r\ddot\psi\cos\theta)\hat a_1 + (\ddot\psi\, d - r\dot\theta^2\cos\theta - r\ddot\theta\sin\theta)\hat a_2 + (r\ddot\theta\cos\theta - r\dot\theta^2\sin\theta)\hat a_3 $$

Term II: $\vec\omega^{A/N} \times {}^N\vec{v}_G$.

$$ \vec\omega^{A/N} \times {}^N\vec{v}_G = \begin{vmatrix} \hat a_1 & \hat a_2 & \hat a_3 \\ 0 & 0 & \dot\psi \\ -r\dot\psi\cos\theta & \dot\psi\, d - r\dot\theta\sin\theta & r\dot\theta\cos\theta \end{vmatrix} $$

$$ \vec\omega^{A/N} \times {}^N\vec{v}_G = (r\dot\theta\dot\psi\sin\theta - d\dot\psi^2)\hat a_1 - r\dot\psi^2\cos\theta\;\hat a_2 $$

Sum. Adding the two terms component by component:

$$ \begin{aligned} &\hat a_1: \quad \big(-r\ddot\psi\cos\theta + r\dot\psi\dot\theta\sin\theta\big) + \big(r\dot\theta\dot\psi\sin\theta - d\dot\psi^2\big) = -d\dot\psi^2 - r\ddot\psi\cos\theta + 2r\dot\theta\dot\psi\sin\theta \\ &\hat a_2: \quad \big(\ddot\psi\, d - r\dot\theta^2\cos\theta - r\ddot\theta\sin\theta\big) - r\dot\psi^2\cos\theta = d\ddot\psi - r\ddot\theta\sin\theta - r\big(\dot\theta^2 + \dot\psi^2\big)\cos\theta \\ &\hat a_3: \quad r\ddot\theta\cos\theta - r\dot\theta^2\sin\theta \end{aligned} $$

$$ \boxed{\begin{aligned} {}^N\vec{a}_G = &\Big(-d\dot\psi^2 - r\ddot\psi\cos\theta + 2r\dot\theta\dot\psi\sin\theta\Big)\hat a_1 \\ &+ \Big(d\ddot\psi - r\ddot\theta\sin\theta - r\big(\dot\theta^2 + \dot\psi^2\big)\cos\theta\Big)\hat a_2 \\ &+ \Big(r\ddot\theta\cos\theta - r\dot\theta^2\sin\theta\Big)\hat a_3 \end{aligned}} $$

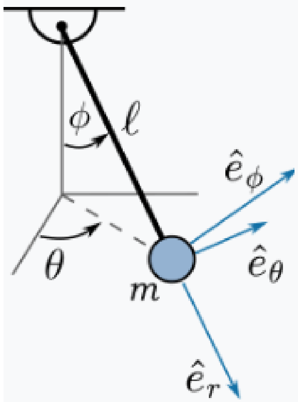
Problem 2

Problem 2

Write the equation of motion of a spherical pendulum of mass m at the end of a rod of length ℓ.

(a) First, derive the acceleration of the spherical pendulum in terms of its rotating reference frame, $(\hat{e}_r, \hat{e}_\theta, \hat{e}_\phi)$. The $\hat{e}_r$ direction points in the pendulum, the θ direction is parallel to the ground (perpendicular to vertical) and points in the direction of increasing θ, and the $\hat{e}_\phi$ direction is perpendicular to both $\hat{e}_r, \hat{e}_\theta$, pointing in the direction of increasing ϕ.

(b) Then, use Newton's 2nd Law to generate the equations of motion, generating 3 equations for the tension in the rod and the angular accelerations $\ddot{\phi}, \ddot{\theta}$.



Setup and Reference Frame

In terms of the fixed basis $(\hat{n}_1, \hat{n}_2, \hat{n}_3)$,

$$\hat{e}_r = \begin{Bmatrix} \sin\phi\cos\theta \\ \sin\phi\sin\theta \\ -\cos\phi \end{Bmatrix}, \qquad \hat{e}_\phi = \begin{Bmatrix} \cos\phi\cos\theta \\ \cos\phi\sin\theta \\ \sin\phi \end{Bmatrix}, \qquad \hat{e}_\theta = \begin{Bmatrix} -\sin\theta \\ \cos\theta \\ 0 \end{Bmatrix}$$

Derivatives of the Rotating Basis

Differentiating each fixed-frame column vector and regrouping gives the kinematic identities

$$\dot{\hat{e}}_r = \dot{\phi}\,\hat{e}_\phi + \dot{\theta}\sin\phi\,\hat{e}_\theta, \qquad \dot{\hat{e}}_\phi = -\dot{\phi}\,\hat{e}_r + \dot{\theta}\cos\phi\,\hat{e}_\theta, \qquad \dot{\hat{e}}_\theta = -\dot{\theta}\sin\phi\,\hat{e}_r - \dot{\theta}\cos\phi\,\hat{e}_\phi$$

Position, Velocity, and Acceleration

With $\vec{r}_{m/O} = \ell\,\hat{e}_r$, differentiating once using $\dot{\hat{e}}_r$ gives the velocity,

$${}^N\vec{v}_m = \ell\dot{\hat{e}}_r = \ell(\dot{\phi}\,\hat{e}_\phi + \dot{\theta}\sin\phi\,\hat{e}_\theta)$$

Differentiating again, and using $\dot{\hat{e}}_\phi$ and $\dot{\hat{e}}_\theta$ to eliminate all derivatives of unit vectors,

$${}^N\vec{a}_m = \ell\Big[\ddot{\phi}\,\hat{e}_\phi + \dot{\phi}\dot{\hat{e}}_\phi + (\ddot{\theta}\sin\phi + \dot{\theta}\dot{\phi}\cos\phi)\hat{e}_\theta + \dot{\theta}\sin\phi\,\dot{\hat{e}}_\theta\Big]$$

Substituting $\dot{\hat{e}}_\phi = -\dot{\phi}\,\hat{e}_r + \dot{\theta}\cos\phi\,\hat{e}_\theta$ and $\dot{\hat{e}}_\theta = -\dot{\theta}\sin\phi\,\hat{e}_r - \dot{\theta}\cos\phi\,\hat{e}_\phi$ and collecting components:

$$\begin{aligned}
\hat{e}_r: &\quad -\dot{\phi}^2 - \dot{\theta}^2 \sin^2\phi \\
\hat{e}_\theta: &\quad \ddot{\theta}\sin\phi + 2\dot{\phi}\dot{\theta}\cos\phi \\
\hat{e}_\phi: &\quad \ddot{\phi} - \dot{\theta}^2\sin\phi\cos\phi
\end{aligned}$$

$$\boxed{{}^N\vec{a}_m = \ell\Big[-(\dot{\phi}^2 + \dot{\theta}^2\sin^2\phi)\hat{e}_r + (\ddot{\theta}\sin\phi + 2\dot{\phi}\dot{\theta}\cos\phi)\hat{e}_\theta + (\ddot{\phi} - \dot{\theta}^2\sin\phi\cos\phi)\hat{e}_\phi\Big]}$$

Equations of Motion (Newton's Second Law)

The forces on the mass are the rod tension, directed along $-\hat{e}_r$, and gravity:

$$\vec{F} = -T\,\hat{e}_r - mg\,\hat{n}_3$$

Solving the basis relations above for $\hat{n}_3$ gives

$$\hat{n}_3 = -\cos\phi\,\hat{e}_r + \sin\phi\,\hat{e}_\phi$$

so that, in the rotating basis,

$$\vec{F} = (mg\cos\phi - T)\hat{e}_r - mg\sin\phi\,\hat{e}_\phi$$

Applying $\vec{F} = m\,{}^N\vec{a}_m$ component by component:

$$\begin{aligned}
\hat{e}_r: &\quad mg\cos\phi - T = -m\ell(\dot{\phi}^2 + \dot{\theta}^2\sin^2\phi) \\
\hat{e}_\theta: &\quad 0 = m\ell(\ddot{\theta}\sin\phi + 2\dot{\phi}\dot{\theta}\cos\phi) \\
\hat{e}_\phi: &\quad -mg\sin\phi = m\ell(\ddot{\phi} - \dot{\theta}^2\sin\phi\cos\phi)
\end{aligned}$$

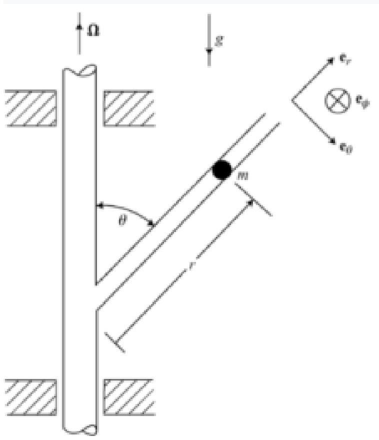
Solving each for the angular accelerations and the tension:

$$\boxed{\begin{aligned}
\ddot{\phi} &= \dot{\theta}^2\sin\phi\cos\phi - \frac{g}{\ell}\sin\phi \\
\ddot{\theta} &= -2\dot{\phi}\dot{\theta}\cot\phi \\
T &= mg\cos\phi + m\ell(\dot{\phi}^2 + \dot{\theta}^2\sin^2\phi)
\end{aligned}}$$

Problem 3

Problem 3

A bead of mass m is constrained to slide without friction in the tube, which is welded to a vertical post at the fixed angle θ. The post rotates with constant angular velocity Ω.



(a) What is the (inertial) acceleration vector expressed in the E-frame shown?

(b) Find the equation of motion for $\ddot{r}$.

(c) Taking $\theta = 60^\circ$, $\Omega = 2$ rad/s, and $\dot{r}_0 = 0$, plot r vs time for a range of initial conditions with $r_0 \in [0.5, 5]$ over the time interval $t \in [0, 1.5]$ seconds. Choose any mass.

(d) For each of those plots, also plot the magnitude of normal force ($F_N = \sqrt{F_{N\phi}^2 + F_{N\theta}^2}$) vs time. *Note: you may also consider plotting these in various colors on the same two plots (r vs. t and F_N vs. t).*

Setup and Reference Frame

The rotating tube frame E has basis $(\hat{e}_r, \hat{e}_\theta, \hat{e}_\phi)$, with $\hat{e}_r$ along the tube axis. Since the tube makes a fixed angle θ with the post,

$$\hat{n}_3 = \cos\theta\,\hat{e}_r - \sin\theta\,\hat{e}_\theta, \qquad \vec{\omega}^{E/N} = \Omega\,\hat{n}_3 = \Omega\big(\cos\theta\,\hat{e}_r - \sin\theta\,\hat{e}_\theta\big)$$

Velocity and Acceleration of the Bead

The bead's position is $\vec{r}_m = r\,\hat{e}_r$. Applying the transport theorem,

$$^N\vec{v}_m = \frac{^E d}{dt}\vec{r}_m + \vec{\omega}^{E/N} \times \vec{r}_m = \dot{r}\,\hat{e}_r + \Omega r\big(\cos\theta\,\hat{e}_r - \sin\theta\,\hat{e}_\theta\big) \times \hat{e}_r$$

Using $\hat{e}_r \times \hat{e}_r = 0$ and $\hat{e}_\theta \times \hat{e}_r = -\hat{e}_\phi$,

$$^N\vec{v}_m = \dot{r}\,\hat{e}_r + r\Omega\sin\theta\,\hat{e}_\phi$$

Differentiating again,

$$^N\vec{a}_m = \frac{^E d}{dt}\big(^N\vec{v}_m\big) + \vec{\omega}^{E/N} \times {}^N\vec{v}_m = \big(\ddot{r}\,\hat{e}_r + \dot{r}\Omega\sin\theta\,\hat{e}_\phi\big) + \Omega\big(\cos\theta\,\hat{e}_r - \sin\theta\,\hat{e}_\theta\big) \times \big(\dot{r}\,\hat{e}_r + r\Omega\sin\theta\,\hat{e}_\phi\big)$$

Using $\hat{e}_r \times \hat{e}_\phi = -\hat{e}_\theta$, $\hat{e}_\theta \times \hat{e}_r = -\hat{e}_\phi$, $\hat{e}_\theta \times \hat{e}_\phi = \hat{e}_r$:

$$\vec{\omega}^{E/N} \times {}^N\vec{v}_m = -r\Omega^2\sin^2\theta\,\hat{e}_r - r\Omega^2\sin\theta\cos\theta\,\hat{e}_\theta + \dot{r}\Omega\sin\theta\,\hat{e}_\phi$$

$$\boxed{^N\vec{a}_m = (\ddot{r} - r\Omega^2\sin^2\theta)\hat{e}_r - r\Omega^2\sin\theta\cos\theta\,\hat{e}_\theta + 2\dot{r}\Omega\sin\theta\,\hat{e}_\phi}$$

Equation of Motion (Newton's Second Law)

The forces on the bead are gravity and the frictionless normal force of the tube wall (no $\hat{e}_r$ component):

$$\vec{F} = -mg\,\hat{n}_3 + F_{N\theta}\,\hat{e}_\theta + F_{N\phi}\,\hat{e}_\phi = (-mg\cos\theta)\hat{e}_r + (mg\sin\theta + F_{N\theta})\hat{e}_\theta + F_{N\phi}\,\hat{e}_\phi$$

Applying $\vec{F} = m\,{}^{N}\vec{a}_m$ component by component:

$$\begin{aligned}
\hat{e}_r:&\quad -mg\cos\theta = m(\ddot{r} - r\Omega^2\sin^2\theta)\\
\hat{e}_\theta:&\quad mg\sin\theta + F_{N\theta} = -mr\Omega^2\sin\theta\cos\theta\\
\hat{e}_\phi:&\quad F_{N\phi} = 2m\dot{r}\Omega\sin\theta
\end{aligned}$$

$$\boxed{\ddot{r} = r\Omega^2\sin^2\theta - g\cos\theta}$$

and the normal-force components used in part (d),

$$F_{N\theta} = -m(g\sin\theta + \Omega^2 r\sin\theta\cos\theta), \qquad F_{N\phi} = 2m\Omega\dot{r}\sin\theta, \qquad F_N = \sqrt{F_{N\theta}^2 + F_{N\phi}^2}$$

Numerical Implementation (Parts c–d)

$\theta = 60°$, $\Omega = 2$ rad/s, $\dot{r}_0 = 0$, $r_0 \in [0.5, 5]$.

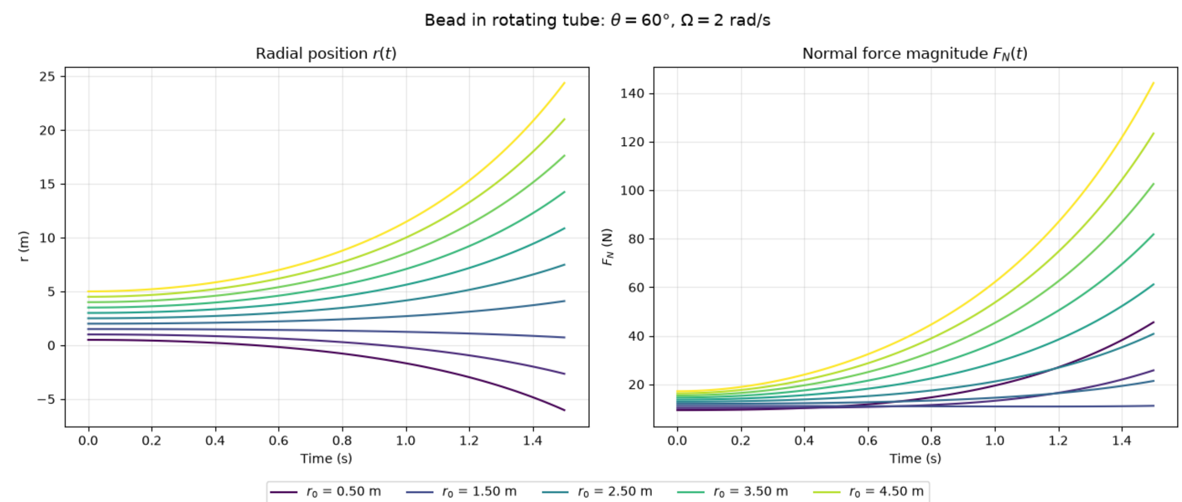
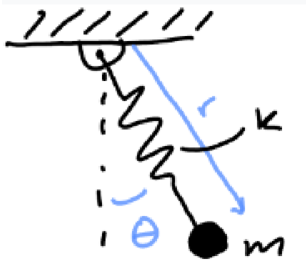


Figure 1: Radial position $r(t)$ (part c) and normal-force magnitude $F_N(t)$ (part d) for $r_0 \in [0.5, 5]$, $\theta = 60°$, $\Omega = 2$ rad/s.

Problem 4

Problem 4

Consider a pendulum where the rod connecting the mass m and its pivot point has some elasticity k. The length of the pendulum at any given time is r, and the counterclockwise angle relative to vertical of the pendulum is θ.



(a) Using r and θ as your coordinates, write the equations of motion for $\ddot{r}$ and $\ddot{\theta}$.

(b) What are the two equilibrium points for this system? The equilibrium points for this system are the pairs of points (r_e, θ_e) for which the velocity and acceleration terms $(\dot{r}, \ddot{r}, \dot{\theta}, \ddot{\theta})$ all go to zero.

(c) Using software such as Python or MATLAB, numerically integrate your equations of motion, and plot the path taken by the mass ($x = r\sin(\theta)$ vs $y = -r\cos(\theta)$). Choose your own values for m, k, r_0 and test with different initial conditions. Try a point very close to each equilibrium point such as $(r_0, \theta_0, \dot{r}_0, \dot{\theta}_0) = (r_e, \theta_e, 0, 0.01)$. Include two or three interesting plots of x vs y.

(d) Write the kinetic and potential energy of the pendulum.

(e) From the results of the numerical integration, plot the kinetic and potential energies as functions of time, as well as the total energy as a function of time.

Setup and Reference Frame

The rotating frame E has basis $(\hat{e}_r, \hat{e}_\theta)$, with $\hat{e}_r$ pointing from the pivot to the mass, so $\vec{\omega}^{E/N} = \dot{\theta}\,\hat{n}_3$, where $\hat{n}_3$ is the fixed out-of-plane axis. Since $\theta = 0$ corresponds to the mass hanging straight down,

$$\hat{n}_{\text{down}} = \cos\theta\,\hat{e}_r - \sin\theta\,\hat{e}_\theta$$

Velocity and Acceleration of the Mass

With $\vec{r}_m = r\,\hat{e}_r$,

$${}^N\vec{v}_m = \frac{{}^E d}{dt}\vec{r}_m + \vec{\omega}^{E/N} \times \vec{r}_m = \dot{r}\,\hat{e}_r + \dot{\theta}\,\hat{n}_3 \times r\hat{e}_r = \dot{r}\,\hat{e}_r + r\dot{\theta}\,\hat{e}_\theta$$

Differentiating again,

$${}^N\vec{a}_m = \frac{{}^E d}{dt}({}^N\vec{v}_m) + \vec{\omega}^{E/N} \times {}^N\vec{v}_m = (\ddot{r}\,\hat{e}_r + \dot{r}\dot{\theta}\,\hat{e}_\theta + r\ddot{\theta}\,\hat{e}_\theta) + \dot{\theta}\,\hat{n}_3 \times (\dot{r}\,\hat{e}_r + r\dot{\theta}\,\hat{e}_\theta)$$

$$\dot{\theta}\,\hat{n}_3 \times (\dot{r}\,\hat{e}_r + r\dot{\theta}\,\hat{e}_\theta) = \dot{r}\dot{\theta}\,\hat{e}_\theta - r\dot{\theta}^2\,\hat{e}_r$$

$$\boxed{{}^N\vec{a}_m = (\ddot{r} - r\dot{\theta}^2)\hat{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{e}_\theta}$$

Part (a): Equations of Motion

The forces on the mass are gravity and the spring tension $T = k(r - r_0)$ directed along $-\hat{e}_r$:

$$\vec{F} = -mg\,\hat{n}_{\text{down}} - T\,\hat{e}_r = (mg\cos\theta - T)\hat{e}_r - mg\sin\theta\,\hat{e}_\theta$$

Applying $\vec{F} = m\,{}^N\vec{a}_m$ component by component:

$$\begin{aligned}\hat{e}_r: &\quad mg\cos\theta - k(r - r_0) = m(\ddot{r} - r\dot{\theta}^2)\\ \hat{e}_\theta: &\quad -mg\sin\theta = m(r\ddot{\theta} + 2\dot{r}\dot{\theta})\end{aligned}$$

$$\begin{aligned}\ddot{r} &= r\dot{\theta}^2 + g\cos\theta - \frac{k}{m}(r - r_0)\\ \ddot{\theta} &= -\frac{g\sin\theta + 2\dot{r}\dot{\theta}}{r}\end{aligned}$$

Part (b): Equilibrium Points

Setting $\dot{r} = \ddot{r} = \dot{\theta} = \ddot{\theta} = 0$ in the equations of motion:

$$0 = -\frac{g\sin\theta}{r} \;\Rightarrow\; \sin\theta = 0 \;\Rightarrow\; \theta_e = 0, \pi$$

$$0 = g\cos\theta - \frac{k}{m}(r - r_0) \;\Rightarrow\; r = r_0 + \frac{mg}{k}\cos\theta$$

$$(r_e, \theta_e) = \left(r_0 + \frac{mg}{k},\ 0\right) \qquad \text{and} \qquad (r_e, \theta_e) = \left(r_0 - \frac{mg}{k},\ \pi\right)$$

Numerical Implementation (Parts c, e)

$m = 1$ kg, $k = 10$ N/m, $r_0 = 1$ m, $\dot{r}_0 = 0$, $\dot{\theta}_0 = 0.01$ rad/s near each equilibrium $\theta_e = 0, \pi$; and $(r, \theta, \dot{r}, \dot{\theta})_0 = (1,\ \pi/2,\ 0.5,\ 1)$ for a large swing away from equilibrium.

Part (c): Trajectory

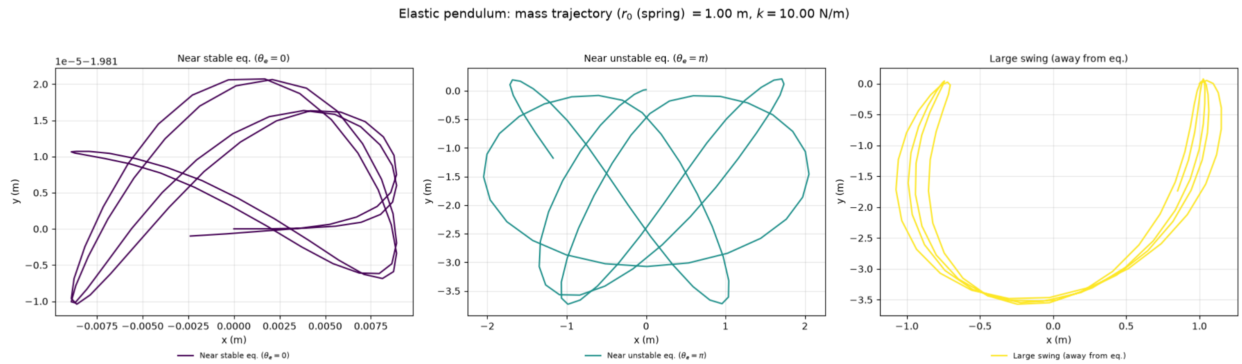


Figure 2: Trajectory (x, y) near the stable equilibrium ($\theta_e = 0$), near the unstable equilibrium ($\theta_e = \pi$), and for a large swing away from equilibrium.

Part (d): Kinetic and Potential Energy

$$^N\vec{v}_m \cdot {}^N\vec{v}_m = \dot{r}^2 + r^2\dot{\theta}^2 \qquad \Longrightarrow$$

$$\boxed{T = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2)}$$

Taking the pivot as the reference height, with $y = -r\cos\theta$ and spring PE $\frac{1}{2}k(r - r_0)^2$,

$$\boxed{V = \frac{1}{2}k(r - r_0)^2 - mgr\cos\theta}$$

Part (e): Energy vs. Time

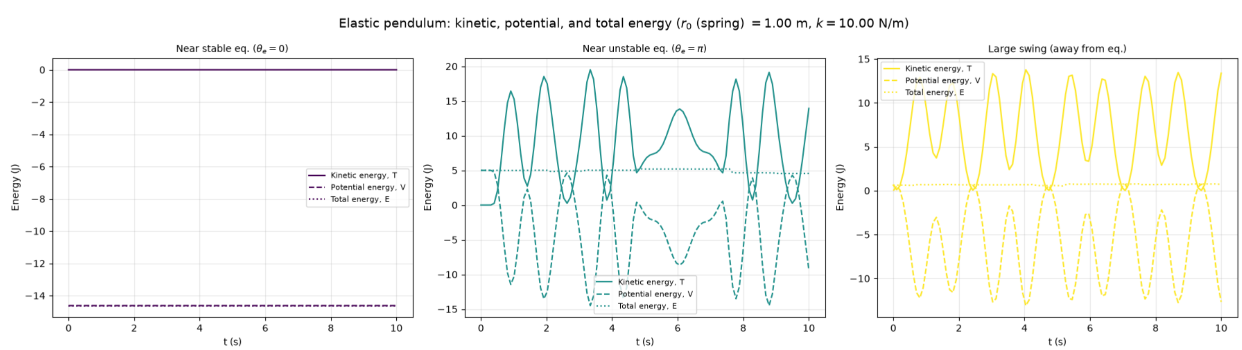


Figure 3: Kinetic, potential, and total energy vs. time for each trajectory of Figure 2.

Appendix: Code Listings

Problem 3 Code

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

theta = np.deg2rad(60)
omega = 2
g = 9.81
r_dot_init = 0
r_span = (0.5 , 5)
t_span =(0,1.5)
m = 1

def system(t, state):
    r, v = state
    drdt = v
    dvdt = r * omega**2 *(np.sin(theta))**2 - g * np.cos(theta)
    return [drdt , dvdt]

t_eval = np.linspace(t_span[0], t_span[-1], 100)
r_eval = np.linspace(r_span[0] , r_span[-1],10)

fig, (ax_r, ax_fn) = plt.subplots(1, 2, figsize=(13, 5.5))
colors = plt.cm.viridis(np.linspace(0, 1, len(r_eval)))

for r, color in zip(r_eval, colors):
    solution = solve_ivp(system, t_span, (r, r_dot_init), t_eval=t_eval)
    t_points = solution.t
    r_values = solution.y[0]
    v_values = solution.y[1]

    F_N_theta = -m * np.sin(theta) * (omega**2 * r_values * np.cos(theta) + g)
    F_N_phi = 2 * m * omega * np.sin(theta) * v_values
    F_N = np.sqrt(F_N_theta**2 + F_N_phi**2)

    label = f'$r_0$ = {r:.2f} m'
    ax_r.plot(t_points, r_values, label=label, color=color)
    ax_fn.plot(t_points, F_N, label=label, color=color)

ax_r.set_xlabel('Time (s)')
ax_r.set_ylabel('r (m)')
ax_r.set_title(r'Radial position $r(t)$')
ax_r.grid(True, alpha=0.3)

ax_fn.set_xlabel('Time (s)')
ax_fn.set_ylabel(r'$F_N$ (N)')
ax_fn.set_title(r'Normal force magnitude $F_N(t)$')
ax_fn.grid(True, alpha=0.3)

handles, labels = ax_r.get_legend_handles_labels()
fig.legend(handles, labels, loc='lower center', ncol=5, bbox_to_anchor=(0.5, -0.05))
fig.suptitle(r'Bead in rotating tube: $\theta = 60°$, $\Omega = 2$ rad/s', fontsize=13)
fig.tight_layout(rect=[0, 0.05, 1, 1])
plt.show()
```

Listing 1: Numerical integration of the rotating-tube bead equation of motion and normal force.

Problem 4 Code

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

g = 9.81
m = 1
k = 10
r_spring_0 = 1
r_dot_0 = 0
omega_0 = .01



t_span =(0,10)
theta_span = (0 , np.pi)

r_span = (r_spring_0 + (m*g) / k , r_spring_0 - (m*g) / k)


def system(t, state):
    r, v , theta, omega = state
    drdt = v
    dthetadt = omega

    dvdt = r * omega**2 + g*np.cos(theta) - (k/m) * (r - r_spring_0)
    domegadt = - ( (g * np.sin(theta) + 2 * v * omega ) / r)


    return [drdt , dvdt , dthetadt , domegadt]

t_eval = np.linspace(t_span[0], t_span[-1], 1000)

cases = [
    dict(label=r'Near stable eq. ($\theta_e=0$)', theta0=theta_span[0], r0=r_span[0], r_dot0=r_dot_0, omega0=omega_0),
    dict(label=r'Near unstable eq. ($\theta_e=\pi$)', theta0=theta_span[1], r0=r_span[1], r_dot0=r_dot_0, omega0=omega_0),
    dict(label='Large swing (away from eq.)', theta0=np.pi / 2, r0=r_spring_0, r_dot0=0.5, omega0=1.0),
]

colors = plt.cm.viridis(np.linspace(0, 1, len(cases)))

fig, axes_r = plt.subplots(1, len(cases), figsize=(6 * len(cases), 6))
fig_E, axes_E = plt.subplots(1, len(cases), figsize=(6 * len(cases), 5))

for case, color, ax_r, ax_E in zip(cases, colors, axes_r, axes_E):
    solution = solve_ivp(system, t_span, (case['r0'], case['r_dot0'], case['theta0'], case['omega0']), t_eval=t_eval)
    t_points = solution.t
    r_values = solution.y[0]
    v_values = solution.y[1]
    theta_values = solution.y[2]
    omega_values = solution.y[3]

    T = (1/2) * m * (v_values**2 + (r_values * omega_values)**2)
    V = (1/2) * k * (r_values - r_spring_0)**2 - m * g * r_values * np.cos(theta_values)
```

```
    E = T + V

    x = r_values * np.sin(theta_values)

    y = - r_values * np.cos(theta_values)

    ax_r.plot(x, y, color=color, label=case['label'])
    ax_r.set_xlabel('x (m)')
    ax_r.set_ylabel('y (m)')
    ax_r.set_title(case['label'], fontsize=10)
    ax_r.grid(True, alpha=0.3)
    ax_r.legend(loc='upper center', bbox_to_anchor=(0.5, -0.12), fontsize=8, frameon=False)

    ax_E.plot(t_points, T, color=color, linestyle='-', label='Kinetic energy, T')
    ax_E.plot(t_points, V, color=color, linestyle='--', label='Potential energy, V')
    ax_E.plot(t_points, E, color=color, linestyle=':', label='Total energy, E')
    ax_E.set_xlabel('t (s)')
    ax_E.set_ylabel('Energy (J)')
    ax_E.set_title(case['label'], fontsize=10)
    ax_E.grid(True, alpha=0.3)
    ax_E.legend(loc='best', fontsize=8)


fig.suptitle(r'Elastic pendulum: mass trajectory ($r_0$ (spring) $= %.2f$ m, $k = %.2f$ N/m)' % (
    r_spring_0, k), fontsize=13)
fig.tight_layout(rect=[0, 0.08, 1, 0.95])

fig_E.suptitle(r'Elastic pendulum: kinetic, potential, and total energy ($r_0$ (spring) $= %.2f$ m,
    $k = %.2f$ N/m)' % (r_spring_0, k), fontsize=13)
fig_E.tight_layout()

plt.show()
```

Listing 2: Numerical integration of the elastic pendulum equations of motion, trajectory, and energy diagnostics.